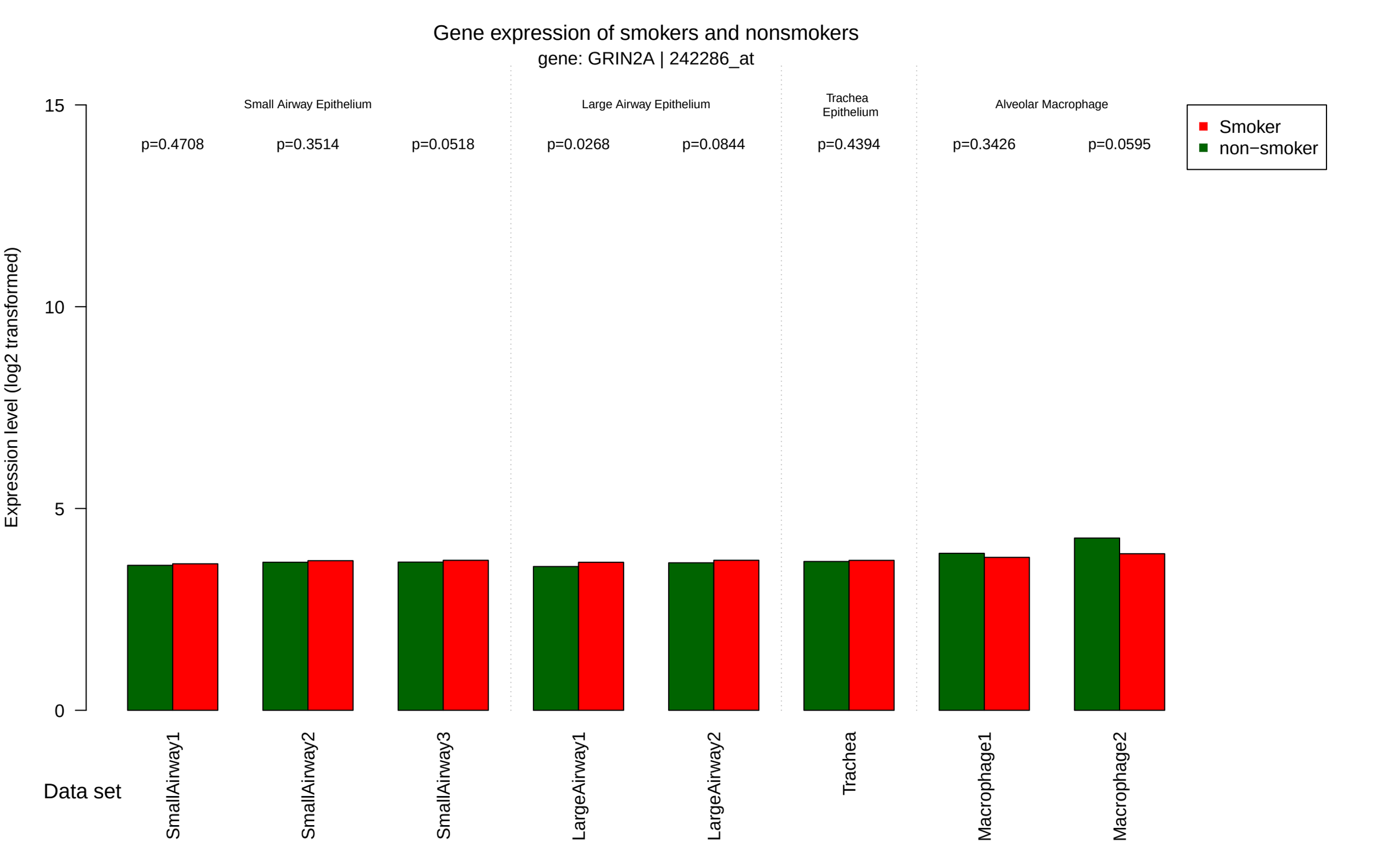
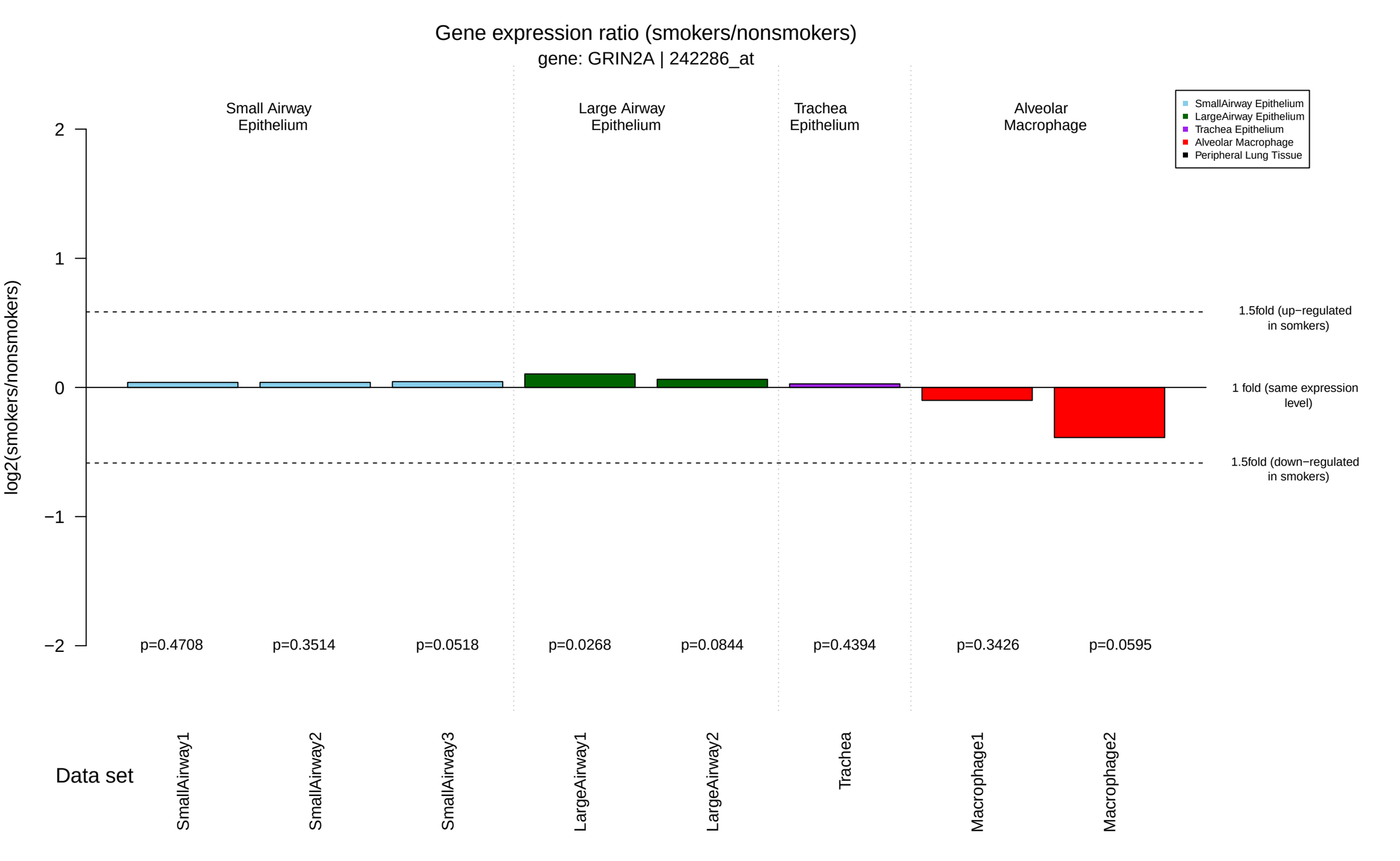
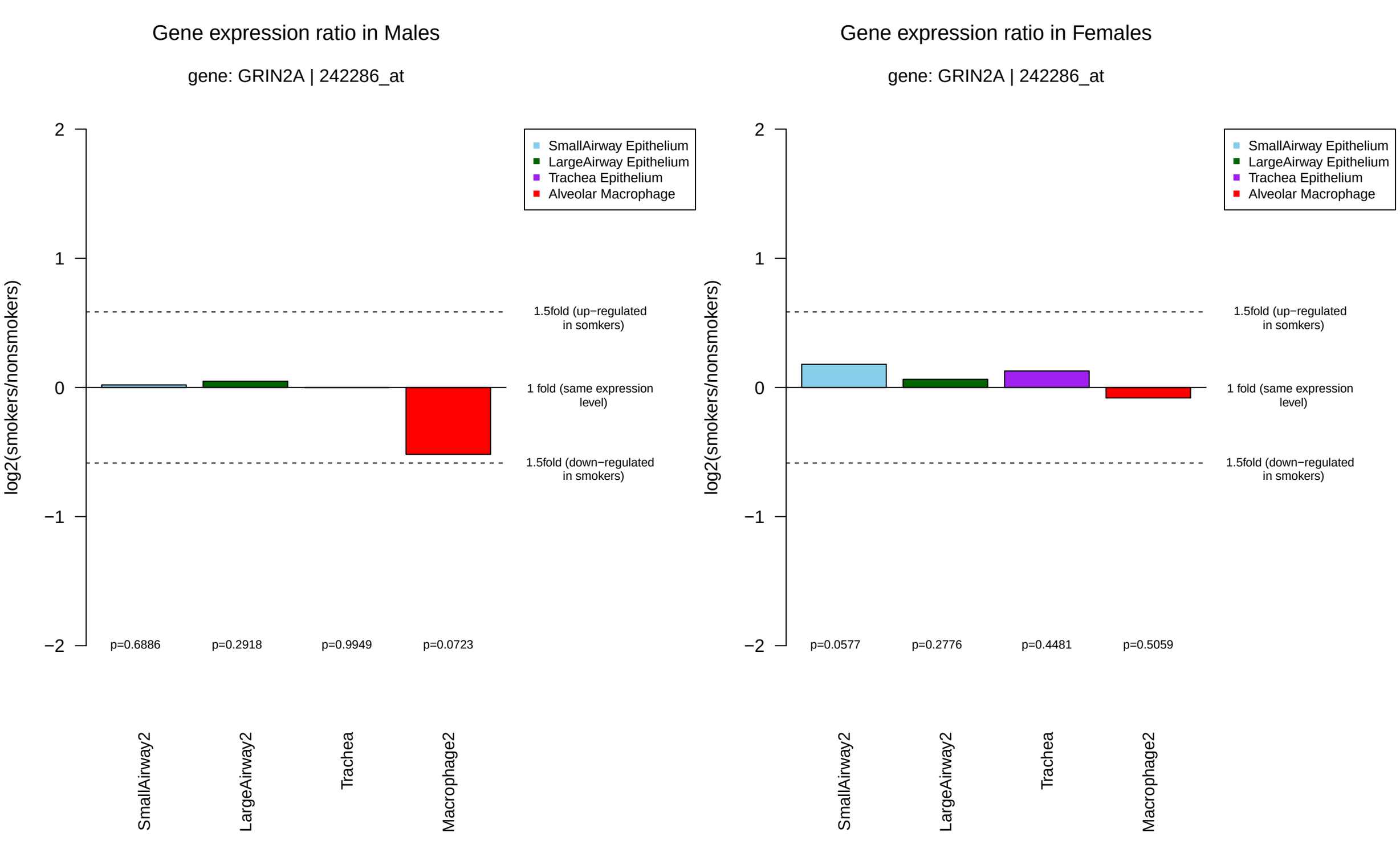
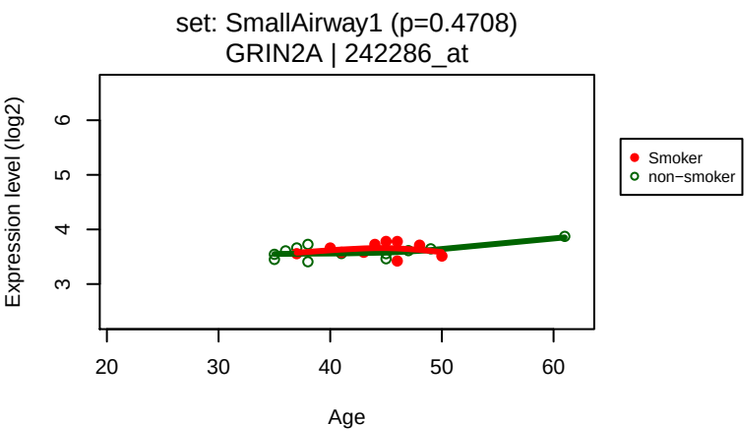
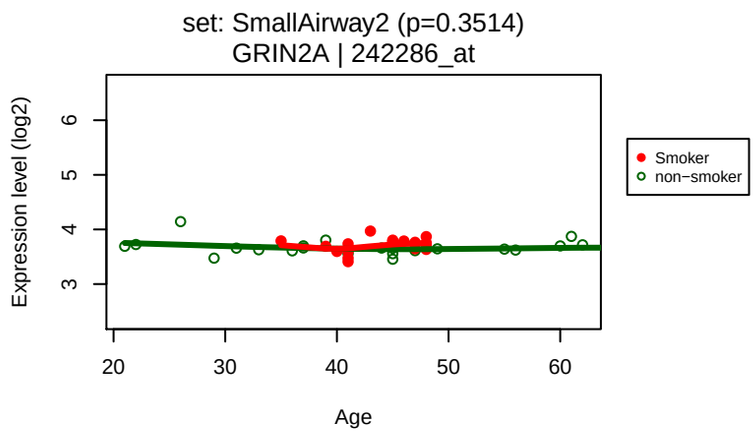
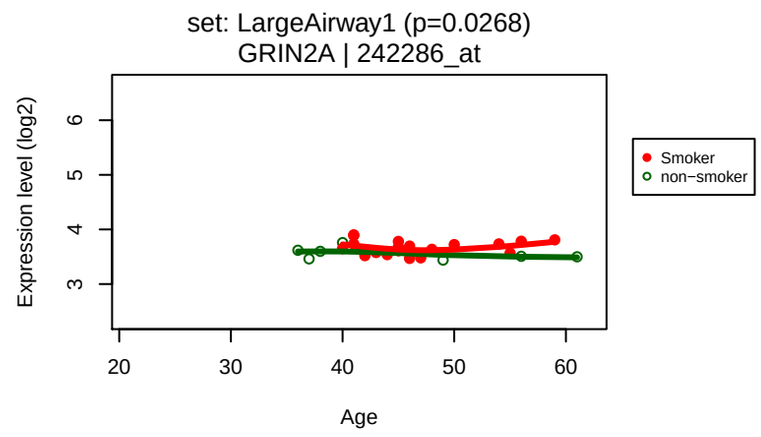

Gene expression of smokers and nonsmokers
gene: GRIN2A | 242286_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4708
p=0.3514
p=0.0518
p=0.0268
p=0.0844
p=0.4394
p=0.3426
p=0.0595
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

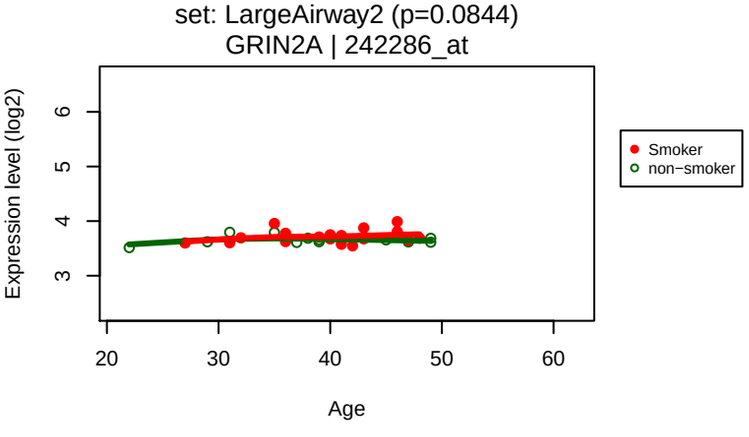

Gene expression ratio (smokers/nonsmokers)
gene: GRIN2A | 242286_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4708
p=0.3514
p=0.0518
p=0.0268
p=0.0844
p=0.4394
p=0.3426
p=0.0595
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

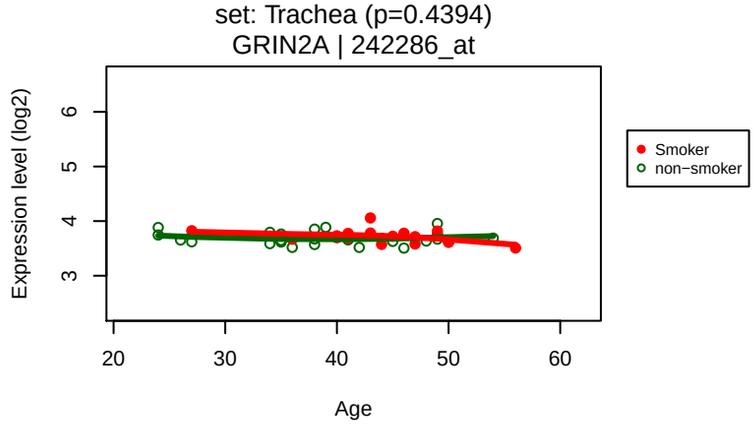

Gene expression ratio in Males
gene: GRIN2A | 242286_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6886
p=0.2918
p=0.9949
p=0.0723
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIN2A | 242286_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0577
p=0.2776
p=0.4481
p=0.5059
SmallAirway2
LargeAirway2
Trachea
Macrophage2

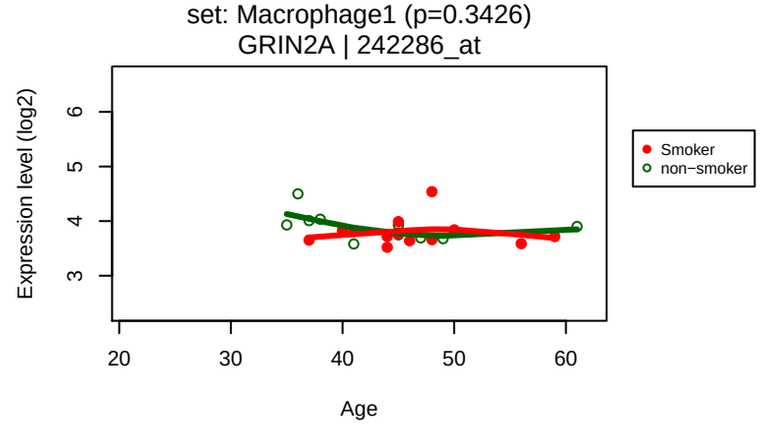

set: SmallAirway1 (p=0.4708)
GRIN2A | 242286_at
Expression level (log2)
Age
Smoker
non−smoker

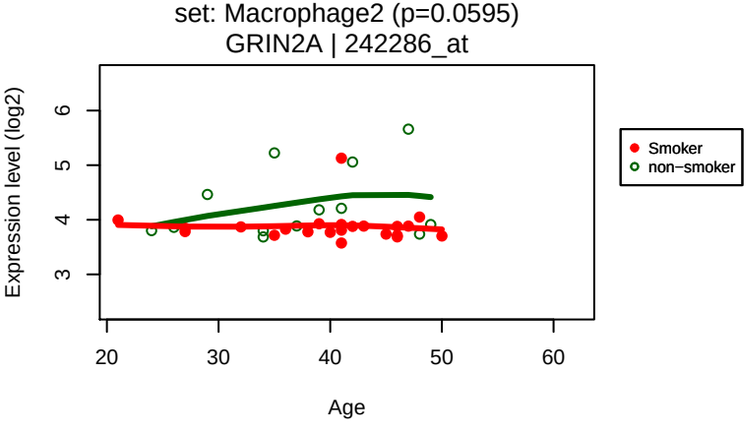

set: SmallAirway2 (p=0.3514)
GRIN2A | 242286_at
Expression level (log2)
Age
Smoker
non−smoker

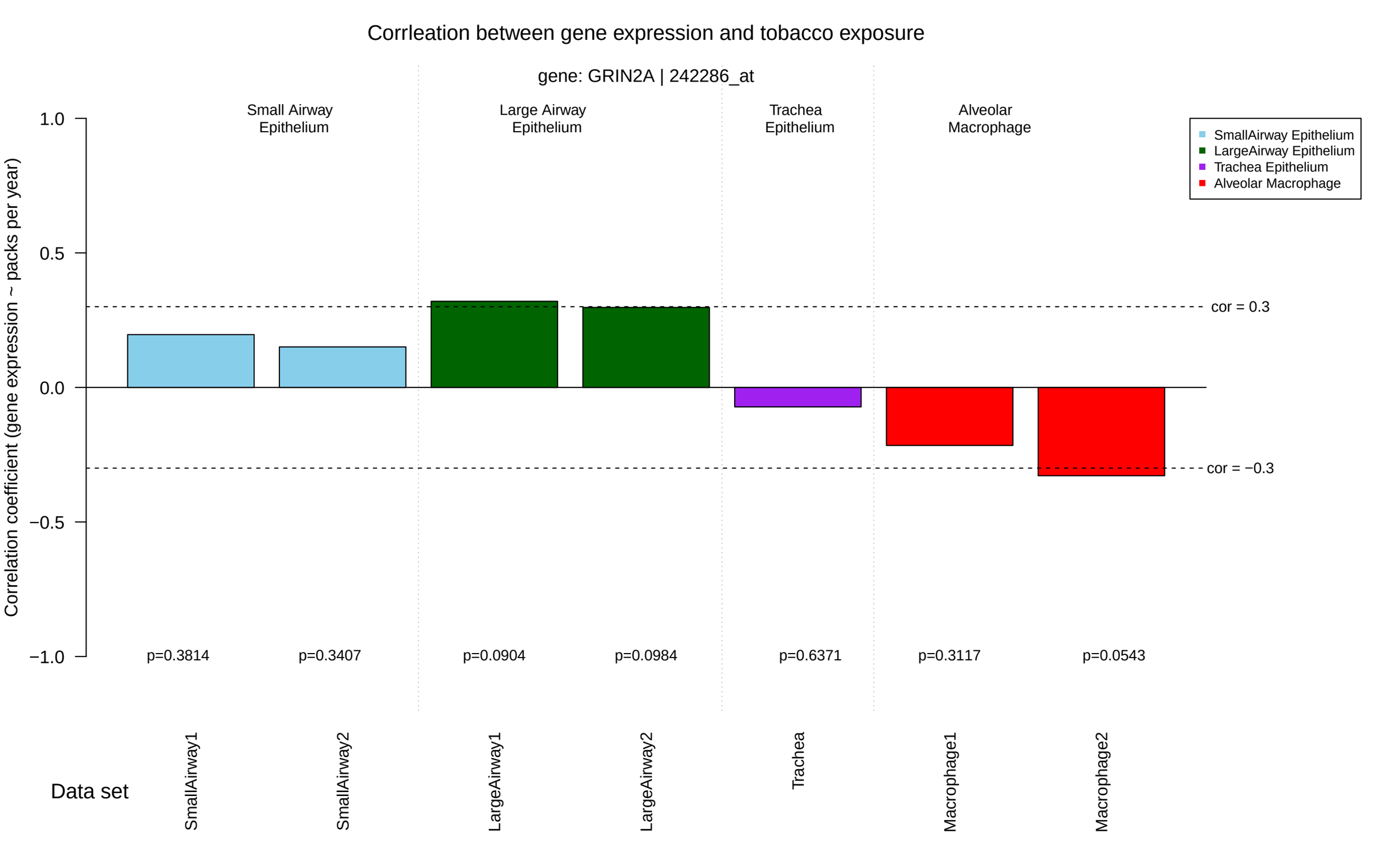

set: LargeAirway1 (p=0.0268)
GRIN2A | 242286_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0844)
GRIN2A | 242286_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4394)
GRIN2A | 242286_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3426)
GRIN2A | 242286_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0595)
GRIN2A | 242286_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIN2A | 242286_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3814
p=0.3407
p=0.0904
p=0.0984
p=0.6371
p=0.3117
p=0.0543
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2